

Week 1 — Course Introduction, Project Map & Setup

SPPH 604 · Application of Advanced Epidemiological Methods

Course orientation

2026-09-10

Welcome to SPPH 604

Application of Advanced Epidemiological Methods

- ▶ An **applied, data-intensive capstone**: epidemiological reasoning through real analysis
- ▶ The goal is to move from theory to **reproducible, defensible practice**
- ▶ We use **just-in-time learning**: methods arrive when your analysis needs them
- ▶ **Today is a one-off orientation**: Thursday, Sept 10 · 10:00–12:00 · Zoom

Land acknowledgement

UBC's Point Grey Campus is located on the **traditional, ancestral, and unceded territory of the x mə k əyəm (Musqueam) people.**

This land has long been a place of learning, knowledge sharing, culture, history, and community.

By the end of today, you should know...

- ▶ how the **term-long project** works
- ▶ how **M0–M4, L1–L6, and P1–P5** fit together
- ▶ what counts as an **eligible project paper**
- ▶ the course rules for **reproducibility, collaboration, and AI use**
- ▶ what to set up before the first regular class/lab week

The big idea: one paper, all term

You choose **one recent, open-data paper** and work with it throughout the course.

Replicate → Review → Improve → Defend → Synthesize

- ▶ **Replicate & interrogate** your own paper
- ▶ **Review** another group's replication
- ▶ **Improve** your own analysis in response to the critique you receive
- ▶ **Defend** what you changed — and what you chose not to change
- ▶ **Synthesize** the final argument in a short reanalysis letter

The regular weekly rhythm

Weeks 2–7

- ▶ **Tuesday 9:30–12:00** · **Zoom** — method + epidemiological reasoning
- ▶ **Thursday 10:00–12:00** · **SPPH 143** — fixed-data lab (L1–L6)
- ▶ **P1–P5** — apply the same idea to your own project

P1–P5 are optional, formative, ungraded, and not separately submitted — no deadline of their own.

They are the building blocks of **M1**, and the practice you draw on at **M2** (critiquing another group's M1) and **M3** (defending your own, alone).

Later in term

From **Weeks 8–12 there is no scheduled Thursday lab**; Tuesday shifts toward presentations, defense, and instructor-supported project work.

The milestone map

Milestone	What happens	When
M0	Paper feasibility gate / lock	Tue Sept 22 · 12 PM
M1	Replication + interrogation	Tue Oct 27 · presentations from 9 AM; deck + code zip due 4 PM
M2	Critique another group's M1	Tue Nov 3 · presentations from 9 AM; slides only due 4 PM
M3	Improved analysis + live individual defense	Tue Nov 17 · from 9 AM
M4	Final reanalysis letter + reproducible repo	Fri Dec 11 · 4 PM

What each milestone asks you to do

- ▶ **M0 — Feasibility:** prove that your chosen paper can support the whole course
- ▶ **M1 — Replicate + interrogate:** rebuild the main result and identify the **dominant threat**
- ▶ **M2 — Review:** test and critique **another group's M1** professionally and constructively
- ▶ **Between M2 and M3 — Improve:** decide which reviewer points to act on in **your own** analysis
- ▶ **M3 — Defend:** explain what you changed, what you rejected, and **why**
- ▶ **M4 — Synthesize:** turn the full argument into a concise reanalysis letter + reproducible archive

How you are graded

Component	Weight	Grading
Labs L1–L6	10%	Complete / Incomplete
P1–P5	0%	Optional formative work
M0	0%	Pass / revise feasibility gate
M1	30%	Weak / Adequate / Exceptional
M2	15%	Weak / Adequate / Exceptional
M3	15%	Weak / Adequate / Exceptional
M4	30%	Weak / Adequate / Exceptional
Bonus	+2% max	Journal-submission pledge

The course rewards traceable reasoning, not merely reproducing the paper's exact number.

Choosing your project: the M0 feasibility gate

A paper qualifies only if your group can point to all six:

1. **Reconstructable public microdata**
2. A clear **primary exposure/predictor and outcome**
3. Plausible **measured confounders** for a defensible adjustment set
4. Enough **observations/events** for the main analysis and at least one subgroup/interaction analysis
5. The **design variables** your analysis has to respect — survey weights/strata/PSU, or follow-up time and censoring
6. A **genuine incomplete-observation problem** — a variable your locked analysis uses that is missing for records which should have carried it

Do not add variables simply to manufacture a missing-data problem.

BREAK — 10 min

Back in 10; then reproducibility, AI, and collaboration.

Reproducibility is the standard

- ▶ Every submitted **number, figure, and table** should be generated by code
- ▶ The pipeline must run **start to finish on a fresh machine** without manual intervention
- ▶ Your group works in a **shared private GitHub repository** during the term
- ▶ **M1 and M4** include a downloaded reproducible repository **zip**
- ▶ **M2 is slides only** — you rerun the assigned group's M1 materials, but submit no new code archive
Code that only works on one person's laptop is unfinished analysis.

AI is a co-pilot — accountability stays with you

- ▶ AI tools are **permitted; none is required**
- ▶ You may use them for appropriate coding, debugging, drafting, and preparation **within the rules of each assessment**
- ▶ You remain responsible for **auditing the code and defending the epidemiological reasoning**
- ▶ M0, M1, M2 and M4 each carry a **cover note**: contributions, external feedback, AI use
 - ▶ final slide or final page of what you submit — plus COVER-NOTE.md in the repo zip at **M1 and M4**
- ▶ **M3 preparation is unrestricted and undisclosed** — rehearse with an AI if you want; there is no M3 submission to disclose it in
- ▶ **Presentation**: use everything you prepared — slides, your own speaker notes, your own code and output
- ▶ **Q&A**: the answer is yours, spoken — no prepared answer read aloud, no help from anyone in any channel, nothing produced during the session (no AI, no live prompting, no messages)
- ▶ Supports approved through your **Academic Accommodation**

M2 has special confidentiality rules

When you review another group's unpublished M1:

- ▶ **Paste what is yours or public; describe what is theirs**
- ▶ Yours or public: your `sessionInfo()`, your install log, package documentation, an error stripped of their code and numbers
- ▶ Theirs: every line of their code, slides, output, logs, results — not a snippet, not a screenshot, not retyped
- ▶ If you cannot ask without quoting their work, **ask the teaching team**
- ▶ AI must **not generate the substantive critique** for you
- ▶ Reviewed material is **confidential** and may not be circulated outside the review assignment
- ▶ Never copy or adapt their code, text, figures, output or distinctive implementation
- ▶ **Credit does not make it permitted** — their solutions stay theirs
- ▶ A general insight may send you back to *your own* analysis: redo it yourself, disclose it

Logistics: what happens where

- ▶ **Today:** Thu Sept 10 · 10:00–12:00 · **Zoom**
- ▶ **Weeks 2–7:** Tue class **Zoom** · Thu lab **in person, SPPH 143**
- ▶ **Weeks 8–12:** no scheduled Thursday lab; Tuesday becomes project/presentation time
- ▶ **Canvas:** Zoom links, weekly modules, rubrics, announcements, and official submission instructions
- ▶ **Learning materials:** two open-access textbooks + free software; nothing to purchase
- ▶ **Communication:** administrative email to the instructor; include “**SPPH 604**” in the subject line
- ▶ Accommodation letters: send to the instructor within the **first two weeks** where applicable

Our demonstration paper (*the only paper slide today*)

- ▶ **Kueh et al., BMJ Open 2026 — MASLD & NHANES 2007–2018**
- ▶ Adults with fatty-liver disease classified into **four body-shape groups** using BMI $\times$ waist-to-height ratio
- ▶ **Headline:** the non-obese/high-waist group shows the poorest survival
- ▶ Across the term, we will **rebuild the analysis from public data and interrogate the choices behind it**
Today is the what and why. The methodological how begins next week and unfolds just in time.

This week: get set up

Before next week's regular class/lab rhythm:

- ▶ Install **R + RStudio + Quarto**
- ▶ Install **Git** and create/access a **GitHub** account
- ▶ Render one small Quarto document successfully
- ▶ Make a first Git commit and push to a **private** test/project repository
- ▶ **Form your group**
- ▶ Begin scanning recent open-data papers for M0

Target for today: leave with the toolchain working — or know exactly what still needs troubleshooting.

Bridge to Week 2

Tuesday Sept 15 · Zoom

- ▶ survey data sources, especially **NHANES**
- ▶ how to read a paper's **Methods** critically
- ▶ feasibility and peer-review principles

Thursday Sept 17 · SPPH 143

- ▶ **L1: data wrangling** on fixed teaching data
- ▶ begin shortlisting papers toward **M0**

Come with: a working toolchain and 1–2 plausible candidate papers.